

Benjamin Hoover

AI Researcher, Engineer, Student

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Education

Machine Learning PhD Student

Aug 2021 - Present

Georgia Tech, Atlanta, GA

Advisor: Duen Horng (Polo) Chau

Master of Engineering: Electrical and Computer Engineering

May 2018

Duke University, Durham, NC

Bachelor of Science in Engineering: Biomedical Engineering

May 2017

Duke University, Durham, NC

Industry Research Experience

IBM Research

Sep '18 - Present

Visual AI Lab, Cambridge, MA

Mentors: Hendrik Strobelt • Dmitry Krotov

Member of the Visual AI Lab where we use visualization to breathe insight and interactivity into powerful AI systems.

Medtronic Diabetes

Summer '17

Algorithms R&D, Northridge, CA

Mentors: Peter Ajemba • Keith Nogueira

Analyze trends in patient blood glucose levels for the sensor R&D team, developing algorithms that were incorporated into product by the end of the summer.

Academic Research Experience

Georgia Institute of Technology

Aug '21 - present

Polo Club at School of Computational Science and Engineering, Atlanta, GA

Mentor: Polo Chau

Member of the Polo Club of Data Science where we innovate scalable, interactive, and interpretable tools that amplify human's ability to understand and interact with billion-scale data and machine learning models.

International Genetically Engineered Machine (iGEM)

Apr '15 - May '17

Lynch Lab, Durham, NC

Mentors: Eirik Adim Moreb • Michael D. Lynch

Designed experiments to analyze CRISPR-Cas9 binding effects in bacteria, using machine learning to discover patterns in the data.

Awards and Honors

Oral Paper Award

July 2025

Concept Attention is an oral presentation (top 1% of 12k+ submissions) at ICML 2025

Best Poster Award Concept Attention is the Best Poster at the CVPR'25 Workshop on Visual Concepts	Jun. 2025
Best Poster Award Transformer Explainer receives Best Poster at IEEE VIS 2024	Oct. 2024
Best Poster Award Energy Transformer receives 2nd place 🏆 in GATech's annual Graduate Poster Symposium	Mar. 2024
Best Paper Award, Honorable Mention DiffusionDB receives a Best Paper, Honorable Mention Award at ACL 2023	July 2023
Spotlight Paper Award HAMUX is spotlighted (top 5 submissions) at the Deep Learning and Differential Equation Workshop at NeurIPS 2022	Dec. 2022
Highlighted Reviewer ICLR 2022	Apr. 2022
Best Poster Award RXNMapper wins Best Poster award at #IOPPoster 2020	July 2020
Best Demo Award LMdiff is Best Demo at NeurIPS 2020	Dec. 2020
(Runner Up) Best Demo exBERT is runner up for Best Demo at NeurIPS 2019	Dec. 2019
Dean's List In the highest one third of engineering class by GPA at Duke	2013-2017
Raul Buelvas Award Given to one freshman at Duke University who "best exemplifies honor, optimism, and selflessness" as voted by peers and faculty	Apr. 2014

Publications

Conference

ConceptAttention: Diffusion Transformers Learn Highly Interpretable Features

Alec Helbling, Tuna Han Salih Meral, **Benjamin Hoover**, Pinar Yanardag, Polo Chau
International Conference on Machine Learning (ICML '25). 2025.

[Code](#) [PDF](#) [Video](#) [Demo](#)

🏆 Oral Spotlight, ICML'25 (top 1%) 🏆 Best Paper, VisCon Workshop@CVPR'25

Dense Associative Memory through the Lens of Random Features

Benjamin Hoover, Polo Chau, Hendrik Strobelt, Parikshit Ram, Dmitry Krotov
Neural Information Processing Systems (NeurIPS). 2024.

[Code](#) [PDF](#) [Poster](#) [Video](#)

Transformer Explainer: Interactive Learning of Text-Generative Models

Aeree Cho, Grace C. Kim, Alexander Karpekov, Alec Helbling, Zijie J. Wang, Seongmin Lee, **Benjamin Hoover**, and Others
IEEE Transactions on Visualization and Computer Graphics (VIS). 2024.

[Demo](#) [PDF](#) [Code](#) [Video](#)

🏆 Best Poster @VIS '24 🏆 Went Viral on X (❤️ 1.9k+) 🏆 #1 paper of the day 🥳

Energy Transformer

Benjamin Hoover, Yuchen Liang, Bao Pham, Rameswar Panda, Hendrik Strobelt, Polo Chau, Mohammed J. Zaki, and Others

Conference on Neural Information Processing Systems (NeurIPS). 2023.

[PDF](#) [Poster](#) [Home](#) [Code](#) [Video \(5m\)](#) [Video \(45m\)](#)

ConceptEvo: Interpreting Concept Evolution in Deep Learning Training

Haekyu Park, Seongmin Lee, **Benjamin Hoover**, Austin Wright, Omar Shaikh, Rahul Duggal, Nilaksh Das, and Others

ACM International Conference on Information and Knowledge Management (CIKM). 2023.

[PDF](#)

Diffusion Explainer: Visual Explanation for Text-to-image Stable Diffusion

Seongmin Lee, **Benjamin Hoover**, Hendrik Strobelt, Zijie J. Wang, ShengYun Peng, Austin Wright, Kevin Li, and Others

IEEE Transactions on Visualization and Computer Graphics (VIS). 2024.

[Demo](#) [PDF](#) [Video](#) [Code](#)

DiffusionDB: A Large-scale Prompt Gallery Dataset for Text-to-Image Generative Models

Zijie J. Wang, Evan Montoya, David Munechika, Haoyang Yang, **Benjamin Hoover**, Polo Chau

Association for Computational Linguistics (ACL). 2023.

[Home](#) [Code](#) [Dataset](#) [PDF](#)

🏆 [Best Paper Award, Honorable Mention](#) 🏆 [Went Viral](#) 🏆 [Oral Presentation](#)

Fairness Evaluation in Text Classification: Machine Learning Practitioner Perspectives of Individual and Group Fairness

Zahra Ashktorab, **Benjamin Hoover**, Mayank Agarwal, Casey Dugan, Werner Geyer, Hao Bang Yang, Mikhail Yurochkin

Conference on Human Factors in Computing Systems (CHI). 2023.

[PDF](#)

Interactive and Visual Prompt Engineering for Ad-hoc Task Adaptation with Large Language Models

Hendrik Strobelt, Albert Webson, Victor Sanh, **Benjamin Hoover**, Johanna Beyer, Hanspeter Pfister, Alexander Rush

IEEE Transactions on Visualization and Computer Graphics (VIS). 2022.

[PDF](#) [Demo](#)

Shared Interest: Measuring Human-AI Alignment to Identify Recurring Patterns in Model Behavior

Angie Boggust, **Benjamin Hoover**, Arvind Satyanarayan, Hendrik Strobelt

ACM Conference on Human Factors in Computing Systems (CHI). 2022.

[Project](#) [Demo](#) [Video](#) [PDF](#) [Code](#)

Can a Fruit Fly Learn Word Embeddings?

Yuchen Liang, Chaitanya K. Ryali, **Benjamin Hoover**, Leopold Grinberg, Saket Navlakha, Mohammed J. Zaki, Dmitry Krotov

International Conference for Learning Representations (ICLR). 2021.

[Project](#) [PDF](#) [Poster](#) [Code](#)

The Design and Development of a Game to Study Backdoor Poisoning Attacks: The Backdoor Game

Zahra Ashktorab, Casey Dugan, James Johnson, Aabhas Sharma, Dustin Ramsey Torres, Ingrid Lange, **Benjamin Hoover**, and Others

International Conference on Intelligent User Interfaces (IUI). 2021.

[PDF](#)

CogMol: Target-Specific and Selective Drug Design for COVID-19 Using Deep Generative Models

Enara Vijil, Payel Das, Samuel Hoffman, Hendrik Strobelt, Inkit Padhi, Kar Wai Lim,

Benjamin Hoover, and Others

Neural Information Processing Systems (NeurIPS). 2020.

[Project](#)

[Demo](#)

[PDF](#)

exBERT: A Visual Analysis of Transformer Models

Benjamin Hoover, Hendrik Strobelt, Sebastian Gehrmann

Association for Computational Linguistics: System Demonstrations (ACL). 2020.

[Project](#)

[Demo](#)

[PDF](#)

[Video](#)

[Code](#)

Journal

Operational response simulation tool for epidemics within refugee and IDP settlements

Joseph Bullock, Carolina Cuesta-Lazaro, Arnau Quera-Bofarull, Anjali Katta, Katherine Hoffmann Pham,

Benjamin Hoover, Hendrik Strobelt, and Others

Public Library of Science: Computational Biology (PLOS). 2021.

[PDF](#)

RXNMapper: Unsupervised Attention Guided Atom Mapping

Philippe Schwaller, **Benjamin Hoover**, JL Reymond, Hendrik Strobelt, Teodoro Laino

Science Advances (AAAS). 2020.

[Project](#)

[Demo](#)

[PDF](#)

[Video](#)

[Code](#)

Managing the SOS Response for Enhanced CRISPR-Cas-Based Recombineering in E. coli through Transient Inhibition of Host RecA Activity

Eirik Adim Moreb, **Benjamin Hoover**, Adam Yaseen, Nisakorn Valyasevi, Zoe Roecker,

Romel Menacho-Melgar, Michael D. Lynch

American Chemical Society: Synthetic Biology (ACS). 2017.

[PDF](#)

Workshop

Dense Associative Memory with Epanechnikov Memory

Benjamin Hoover, Krishna Balasubramanian, Dmitry Krotov, Parikshit Ram

New Frontiers for Associative Memory Workshop at ICLR (NFAM@ICLR). 2025.

[PDF](#)

Memory in Plain Sight: A Survey of the Uncanny Resemblances between Diffusion Models and Associative Memories

Benjamin Hoover, Hendrik Strobelt, Dmitry Krotov, Judy Hoffman, Zsolt Kira, Polo Chau

NeurIPS Workshop on Associative Memory and Hopfield Networks (AMHN@NeurIPS). 2023.

[PDF](#)

[Poster](#)

[Home](#)

HAMUX: A Universal Abstraction for Hierarchical Hopfield Networks

Benjamin Hoover, Polo Chau, Hendrik Strobelt, Dmitry Krotov

Symbiosis of Deep Learning and Differential Equations II Workshop at NeurIPS (DLDE). 2022.

[Code](#)

[Project](#)

[PDF](#)

[Video](#)

Preprints

Tutorial on Modern Methods in Associative Memory

Dmitry Krotov, **Benjamin Hoover**, Parikshit Ram, Bao Pham
International Conference on Machine Learning (ICML '25). 2025.

[Code](#) [PDF](#) [Home](#)

Fairytailor: A multimodal generative framework for storytelling

Eden Bensaid, Mauro Martino, **Benjamin Hoover**, Hendrik Strobelt
ArXiv preprint (ArXiv). 2021.

[PDF](#) [Demo](#)

Talks

Modern Methods in Associative Memory

ICML 2025 Tutorial [Invited](#)

July 2025

Hopfield Networks 2.0: Associative Memory for the Modern Era of AI

Physics Informed ML Workshop at Los Alamos National Laboratory [Invited](#)
Plectics Lab Colloquium [Invited](#)

Sep 2024

May 2024

Energy Transformer

McMahon Lab at Cornell University [Invited](#)

Sep 2023

HAMUX: Inventing Hopfield Networks 2.0

Georgia Tech CSE Hotseat

Nov 2022

Comparing Language Models with LMDiff

Demo at NeurIPS 2020

Dec 2020

exBERT: Exploring Learned Embeddings in Transformer Models

System Demonstration at ACL 2020

Jul 2020

IBM Exposition Demo at ICLR 2020

Apr 2020

System Demonstration at NeurIPS 2019

Dec 2019

Unsupervised Attention Guided Atom Mapping with RXNMapper

ML Interpretability for Scientific Discovery Workshop at ICML 2020

Jul 2020

IBM Exposition Demo at ICML 2020

Jul 2020

Breaking the Wall of Black Box AI

Falling Walls Lab Boston"

Oct 2020

The Inside Story

MEng Graduation elected speaker [Invited](#)

May 2018

Using Matlab's PRT to Accelerate Algorithm Development

Medtronic Diabetes R&D

Jul 2017

The Impact of Encouragement

Cary Christian School Valedictorian Speech [Invited](#)

May 2013

Teaching

Graduate Teaching Assistant

Fall '23 - Spring '24

Georgia Institute of Technology, Atlanta, GA

Instructors: Polo Chau • Mahdi Roozbahani

Organized project teams, graded projects, and held weekly office hours for an online graduate course with 947 students enrolled.

Graduate Teaching Assistant

Spring '18

Duke University, Durham, NC

Instructor: Patrick Wang

| Designed homeworks, supervised projects, and occasionally teach lectures.

Graduate Teaching Assistant

Spring '18

Duke University, Durham, NC

Instructor: Jonathan Viventi

| Supervised and graded capstone projects where students designed and assembled 2-channel EEG readers

Graduate Teaching Assistant

Spring '18

Duke University, Durham, NC

Instructors: Stacy Tantum · Paul Bendich

| As a TA, I helped setup class infrastructure (Python, jupyter) and initial curriculum for this 25 student pilot course for Duke's Data+ Program.

Undergraduate Teaching Assistant

Spring '17

Duke University, Durham, NC

Instructor: Jonathan Viventi

| Prepared, supervised, and graded labs where students designed medical devices such as Ultrasound, ECGs, Baby Incubators, and functional FitBits.

Undergraduate Teaching Assistant

Jan '15 - May '18

Duke University, Durham, NC

Instructor: Stacy Tantum

| Graded homeworks, held weekly office hours, supervised labs, and private tutored students.

Boeing Fellow

Jan '16 - May '17

Duke University, Durham, NC

Instructor: Carmen Rawls

| Wrote STEM lesson plans to engage middle and high school students in the Durham area.

Service

Reviewer

Neural Information Processing Systems (**NeurIPS**) 2024 · 2023 · 2021

International Conference for Learning Representations (**ICLR**) 2025 · 2024 · 2023 · 2022

International Conference for Machine Learning (**ICML**) 2025 · 2023 · 2022

ACM Conference on Human Factors in Computing Systems (**CHI**) 2023

Workshop on Associative Memory & Hopfield Networks at NeurIPS (**AMHN@NeurIPS**) 2023

Workshop on New Frontiers in Associative Memories at ICLR (**NFAM@ICLR**) 2025

Workshop on the Symbiosis of Deep Learning and Differential Equations at NeurIPS (**DLDE@NeurIPS**) 2022

Blogpost Track at ICLR (**BP@ICLR**) 2023

Organizing Committee

Workshop on Associative Memory & Hopfield Network at NeurIPS (**AMHN@NeurIPS**) 2023

Workshop on New Frontiers in Associative Memory at ICLR (**NFAM@ICLR**) 2025